

Medical Accessibility: AI in Dermatology

Rubric

DS 4002

Due: TBD

Submission format: Upload link of github repo to canvas

Individual Assignment

Why am I doing this? The case study will allow you to explore predictive deep learning models with image and multimodal based data. The goal is to expose you to image based data sets and to use data analysis in a way that can tackle real world issues and situations. Given accessibility to healthcare is a big obstacle for many people in the world, this case will allow you to take charge and explore a high stakes real world situation.

What am I going to do? For this case study, you will be given data regarding pigmented skin lesions that have not only the images but also metadata attached (contains some demographic info as well as additional information). To do this you will use code scripts and data provided in the GitHub repository linked here: [HERE](#). After running your code, you will be asked to submit a Github repo that includes the data used, source code, and outputs generated (like EDA, classification reports, and confusion matrices). Finally, you will write a reflection regarding whether multimodal or image only models generated more accurate predictions and the implications of your findings. To summarize, the deliverables include:

- Output folder with all figures created to assist in EDA as well as identify model success and decisions for which parts of the metadata to use.
- A one page reflection regarding the findings and implications of your case study.
- Github Repository that contains all materials used and produced

Tips for success:

- Make sure to read and understand the reference materials.
- Read instructions carefully, but also explore how changing different variables can impact your results. (HINT: Keep track of what you tried and what you learned)
- Talk to your professors or other instructors if you run into any issues.

How will I know I have Succeeded? You will meet expectations on this Case Study when you follow the criteria in the rubric below.

Formatting	● One GitHub Repository containing all materials ○ README.md (auto displays) ○ LICENSE (use MIT as default) ○ Data folder ○ Scripts Folder ○ Output Folder ○ Reflection PDF
README.md	● <u>Goal</u>: This file serves as an orientation to everyone who comes to your repository, it should enable them to get their bearings. ● Use markdown headers to divide content. ● Make an H2 (##) section explaining the contents of the repository ● Section 1: Software and platform section ○ The type(s) of software you used for the project. ○ The names of any add-on packages that need to be installed with the software. ○ The platform (e.g., Windows, Mac, or Linux) you used. ● Section 2: A Map of your documentation. ○ In this section, you should provide an outline or tree illustrating the hierarchy of folders and subfolders contained in your Project Folder, and listing the files stored in each folder or subfolder.
License.md	● <u>Goal</u>: This file explains to a visitor the terms under which they may use and cite your repository. ● Select an appropriate license from the GitHub options list on repository creation. ● Usually, the MIT license is appropriate.
Scripts Folder	● Goal: This folder contains all the source code for your project. ● Include all the scripts you used. Try to name each script according to the order it needs to be executed to reproduce the results.
Data Folder	● Goal: This folder contains all of the data for this project. ● You should AT LEAST include the initial data, and the final data analyzed. If needed, the code in the SCRIPTS folder should be able to get you from the initial piece of data to the final one. N.B. If the initial and final data are the same, then just include that dataset. ● If your data fits in github, place all of it here. ● If your data does not fit in GitHub use a single file explaining the process to obtain the dataset.

Outputs Folder	● <u>Goal</u>: This folder contains all of the output generated by your project, e.g. figures, tables, etc. ● Use subfolders to organize what was created for the EDA as well as confusion matrices and classification reports.
Reflection PDF	● Goal: Reflect on your findings and explain which findings are better and the pros and cons of using an image only vs multimodal approach. ● One page PDF files ● Think about the time it takes to run each model as well as accessibility to information depending on the current state of the patient (ex: can't ID a patient)? ● What would being able to perfect these prediction models mean for the future of healthcare?

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